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Grey-level transformation for a dark museum image:

Technique: Gamma transformation

Procedure:

* Read the input image and convert it to grayscale.
* Normalise the pixel intensity values to the range 0 to 1.
* Apply the gamma transformation formula:

$S = cr^{\gamma}$, where r is the i/p pixel intensity, s is the output intensity, c is a constant.

* Select $\gamma < 1$, such as 0.5, to brighten dark regions.
* Convert the output back to the 8-bit intensity range of 0-255.
* Display and compare the original and enhanced images.

Result: The dark regions become brighter, hidden details of the museum exhibits becomes visible.

2) Fingerprint enhancement using Histogram equalisation:

Technique: Histogram equalisation

Procedure:

* Read the noisy image.
* Select a 3x3 median filter window.
* Move the window over each pixel in the image.
* Sort the neighbouring pixel values and replace the enter pixel.
* Repeat the process for the entire image.
* Display the filtered image and compare it with the noisy image.

Justification: A median filter removes isolated black & white noise pixels while preserving edges better than an averaging filter, which tends to blur boundaries.

Result: Noise is reduced, and important object edges are preserved

3) Salt and pepper noise removal :

Technique : Median filter

Procedure :

* Read the fingerprint image and convert it to grayscale.
* Calculate the histogram to determine the frequency of each intensity level.
* Calculate the cumulative distribution function from histogram.
* Obtain the intensity mapping using:

$$s_k = (L-1) \sum_{j=0}^{k} (p(r_j))$$, where L is the number of intensity and $p(r_j)$ is the probability of each intensity.

* Map the original pixel values to the new intensity values using the CDF.
* Display the enhanced fingerprint image.

Result: The contrast between ridges and valleys increases, improving fingerprint visibility for feature extraction and recognition.

4) PCB track enhancement using image sharpening :

Technique : Laplacian sharpening.

Procedure :

* Read the PCB image and convert it to grayscale.
* Apply a laplacian operator to detect rapid intensity changes.
* Calculate the second order derivative of the image.
* Subtract the laplacian result from the original image.

$$g(x,y) = f(x,y) - \nabla^2 f(x,y)$$

* Adjust the sharpening strength to improve trace visibility.
* Display the sharpened image & compare it with the original.

Result: Copper tracks and circuit edges becomes sharper, making PCB inspection easier.

Drone image smoothing using a frequency domain filter:

Technique: Gaussian low pass filter.

Procedure:

* Read the drone image and convert it to grayscale.
* Apply the fourier transform to convert the image from the spatial domain to the frequency domain.
* Design a gaussian low pass filter:

$$H(U,V) = e^{-D^2(U,V)/(2D_0^2)}$$

* Multiply the fourier transform of the image by the filter.
* Apply the inverse fourier transform to obtain the smoothed image.
* Display the filtered image.

Result: High frequency noise is reduced, making farm-plot outlines easier to observe.

b) Retinal image enhancement using a High-Pass filter

Technique: High-Boost filtering

Procedure:

* Read the retinal fundus image and convert it to grayscale.
* Apply the fourier transform to obtain the frequency-domain representation.
* Design a high-pass filter to emphasize high-frequency components corresponding to fine details and edges.
* Multiply the transformed image by the filter
* Apply the inverse fourier transform to obtain the enhanced image, use highboost filtering:

$$g(x,y) = A f(x,y) - f_{blur}(x,y)$$

Result: Thin blood vessel boundaries and fine structures become more visible. Excessive enhancement should be avoided because it may amplify noise.

7) Point and line detection for Bottle Inspection:

Technique: Point Detection and directional line detection masks

Procedure:

* Read the bottle image and convert it to grayscale
  Apply a point detection mask:

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

* Calculate the mask response to identify isolated intensity changes representing tiny bubbles.
* Apply directional line masks to detect horizontal, vertical, diagonal cracks.
* Compare the absolute filter response with a selected threshold.
* Mark the detected points and lines on the original image.

Result: Tiny air bubbles and hairline cracks are detected for automated bottle quality inspection.

8) Handwriting Extraction using adaptive thresholding:

Technique: adaptive thresholding

Procedure:

* Read the answer sheet image and convert it to grayscale.
* Divide the image into small local neighbourhoods.
* Calculate a threshold for each neighbourhood using the local mean or weighted mean.
* Compare each pixel with its local threshold.
* Convert dark ink pixels black and background pixels to white.
* Display the resulting binary image.

Result: Handwritten text is separated from the background, making it suitable for subsequent character recognition.

1) Lake extraction using region growing:

Technique: Region growing segmentation

Procedure:

* Read the satellite image and select a seed point inside the lake.
* Determine the intensity or color value of the seed pixel
* Examine the neighboring pixels
* Add neighboring pixels that satisfy the similarity condition.

$|I(x,y) - \mu_R| < T$, where $I(x,y)$ is the pixel intensity, μ_R is the current region mean, T is the threshold.

* Update the region statistics as new pixels are added.
* Continue until the no neighboring pixels satisfy the condition.
* Display the segmented lake region.

Result: Discontinuous edges are connected, and a more complete gear outline is obtained for shape analysis and mechanical inspection.

10) Gear outline recovery using edge linking:

Technique: Canny edge detection, edge linking and Boundary detection.

Procedure:

* Read the gear image and convert it to grayscale.
* Apply gaussian smoothing to reduce noise.
* Apply the canny edge detector to identify initial edge pixels.
* Link broken edge segments using local neighborhood connectivity and gradient direction.
* Apply morphological closing to connect small gaps between

nearby edges.

* use contour tracing to identify and extract the gear boundary.
* Display the complete gear outline over the original image.

Result: Discontinuous edges are connected, and a more complete gear outline is obtained for shape analysis and mechanical inspection